

A Practical Guide to Part of speech tagging

TOPIC -- PART OF SPEECH TAGGING

-- 01 --

In corpus linguistics, part-of-speech tagging (POS tagging, PoS tagging, or POST), also called grammatical tagging, is the process of marking up a word in a text (corpus) as corresponding to a particular part of speech, based on both its definition and its context. A simplified form of this is commonly taught to school-age children, in the identification of words as nouns, verbs, adjectives, adverbs, etc. Once performed by hand, POS tagging is now done in the context of computational linguistics, using algorithms which associate discrete terms, as well as hidden parts of speech, by a set of descriptive tags. POS-tagging algorithms fall into two distinctive groups: rule-based and stochastic. E. Brill's tagger, one of the first and most widely used English POS taggers, employs rule-based algorithms.

-- 02 --

The methods already discussed involve working from a pre-existing corpus to learn tag probabilities. It is, however, also possible to bootstrap using "unsupervised" tagging. Unsupervised tagging techniques use an untagged corpus for their training data and produce the tagset by induction. That is, they observe patterns in word use, and derive part-of-speech categories themselves. For example, statistics readily reveal that "the", "a", and "an" occur in similar contexts, while "eat" occurs in very different ones. With sufficient iteration, similarity classes of words emerge that are remarkably similar to those human linguists would expect; and the differences themselves sometimes suggest valuable new insights. These two categories can be further subdivided into rule-based, stochastic, and neural approaches.

-- 03 --

Part-of-speech tagging is harder than just having a list of words and their parts of speech, because some words can represent more than one part of speech at different times, and because some parts of speech are complex. This is not rare—in natural languages (as opposed to many artificial languages), a large percentage of word-forms are ambiguous. For example, even "dogs", which is usually thought of as just a plural noun, can also be a verb:

-- 04 --

Dynamic programming methods In 1987, Steven DeRose and Kenneth W. Church independently developed dynamic programming algorithms to solve the same problem in vastly less time. Their methods were similar to the Viterbi algorithm known for some time in other fields. DeRose used a table of pairs, while Church used a table of triples and a method of estimating the values for triples that were rare or nonexistent in the Brown Corpus (an actual measurement of triple probabilities would require a much larger corpus). Both methods achieved an accuracy of over 95%. DeRose's 1990 dissertation at Brown University included analyses of the specific error types, probabilities, and other related data, and replicated his work for Greek, where it proved similarly effective. These findings were surprisingly disruptive to the field of natural language processing. The accuracy reported was higher than the typical accuracy of very sophisticated algorithms that integrated part of speech choice with many higher levels of linguistic analysis: syntax, morphology, semantics, and so on. CLAWS, DeRose's and Church's methods did fail for some of the known cases where semantics is required, but those proved negligibly rare. This convinced many in the field that part-of-speech tagging could usefully be separated from the other levels of processing; this, in turn, simplified the theory and practice of computerized language analysis and encouraged researchers to find ways to separate other pieces as well. Markov Models became the standard method for the part-of-speech assignment.

-- 05 --

Tag sets Schools commonly teach that there are 9 parts of speech in English: noun, verb, article, adjective, preposition, pronoun, adverb, conjunction, and interjection. However, there are clearly many more categories and sub-categories. For nouns, the plural, possessive, and singular forms can be distinguished. In many languages words are also marked for their "case" (role as subject, object, etc.), grammatical gender, and so on; while verbs are marked for tense, aspect, and other things. In some tagging systems, different inflections of the same root word will get different parts of speech, resulting in a large number of tags. For example, NN for singular common nouns, NNS for plural common nouns, NP for singular proper nouns (see the POS tags used in the Brown Corpus). Other tagging systems use a smaller number of tags and ignore fine differences or model them as features somewhat independent from part-of-speech. In part-of-speech tagging by computer, it is typical to distinguish from 50 to 150 separate parts of speech for English. Work on stochastic methods for tagging Koine Greek (DeRose 1990) has used over 1,000 parts of speech and found that about as many words were ambiguous in that language as in English. A morphosyntactic descriptor in the case of morphologically rich languages is commonly expressed using very short mnemonics, such as

Ncmsan for Category = Noun, Type = common, Gender = masculine, Number = singular, Case = accusative, Animate = no. The most popular "tag set" for POS tagging for American English is probably the Penn tag set, developed in the Penn Treebank project. It is largely similar to the earlier Brown Corpus and LOB Corpus tag sets, though much smaller. In Europe, tag sets from the Eagles Guidelines see wide use and include versions for multiple languages. POS tagging work has been done in a variety of languages, and the set of POS tags used varies greatly with language. Tags usually are designed to include overt morphological distinctions, although this leads to inconsistencies such as case-marking for pronouns but not nouns in English, and much larger cross-language differences. The tag sets for heavily inflected languages such as Greek and Latin can be very large; tagging words in agglutinative languages such as Inuit languages may be virtually impossible. At the other extreme, Petrov et al. have proposed a "universal" tag set, with 12 categories (for example, no subtypes of nouns, verbs, punctuation, and so on). Whether a very small set of very broad tags or a much larger set of more precise ones is preferable, depends on the purpose at hand. Automatic tagging is easier on smaller tag-sets.

-- 06 --

Works cited Charniak, Eugene. 1997. "Statistical Techniques for Natural Language Parsing". *AI Magazine* 18(4):33–44. Hans van Halteren, Jakub Zavrel, Walter Daelemans. 2001. Improving Accuracy in NLP Through Combination of Machine Learning Systems. *Computational Linguistics*. 27(2): 199–229. PDF DeRose, Steven J. 1990. "Stochastic Methods for Resolution of Grammatical Category Ambiguity in Inflected and Uninflected Languages." Ph.D. Dissertation. Providence, RI: Brown University Department of Cognitive and Linguistic Sciences. Electronic Edition available at [1] Archived 2018-08-19 at the Wayback Machine D.Q. Nguyen, D.Q. Nguyen, D.D. Pham and S.B. Pham (2016). "A Robust Transformation-Based Learning Approach Using Ripple Down Rules for Part-Of-Speech Tagging." *AI Communications*, vol. 29, no. 3, pages 409–422. [.pdf]